

國立
臺北科技大學

NATIONAL TAIPEI UNIVERSITY OF TECHNOLOGY



電機系實習報告

課程名稱: 電力電子學實習

實習單元: 直流至直流轉換

單元名稱: 降壓、升壓型電路

班級: 電機三丙

組別: 第十二組

學號及姓名: B812111001 張博翔

111820026 王奕傑

中 華 民 國 114 年 6 月 6 日

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一、電路模擬作業

實習三：直流至直流轉換

--降壓及升壓型轉換器(開迴路) 日期： 114 年 3 月 14 日

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姓名：張博翔 學號：B812111001

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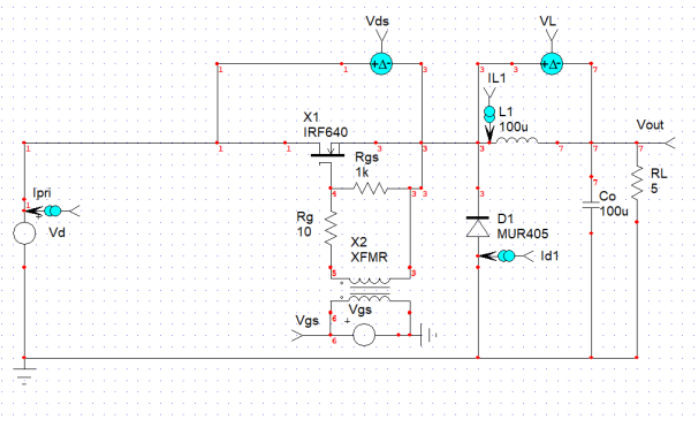
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練習	評 語	評 分
3-1(3)	答案錯誤，-5	95

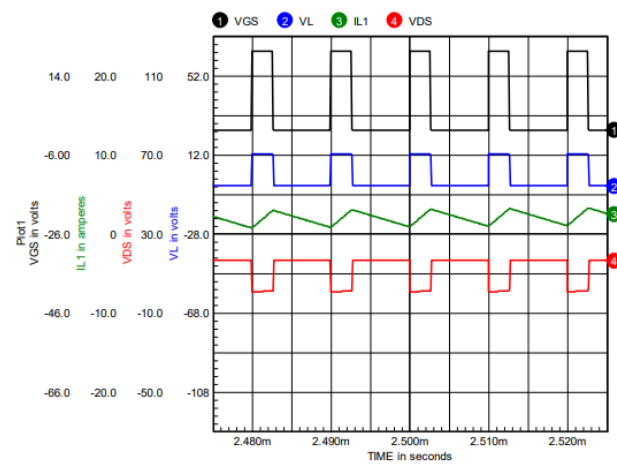
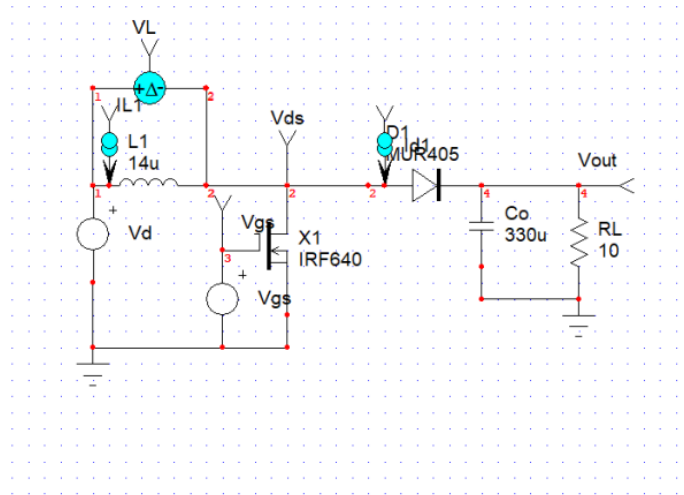
註：每錯一項敘述或每錯一條波形各扣5分。

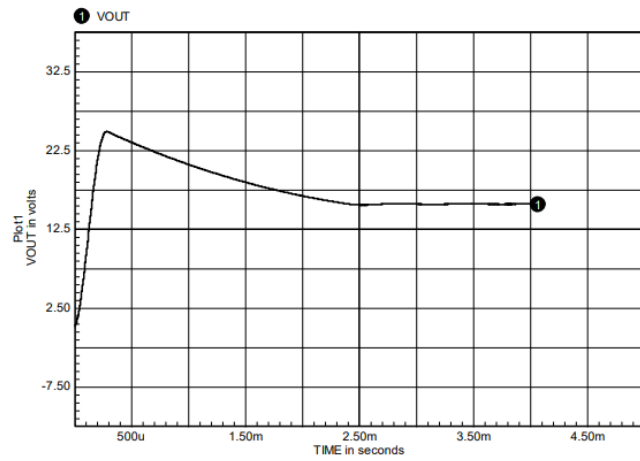
問題與練習：

- 3-1. 實習三 直流至直流轉換電路--降壓型轉換器 Step-Down Converter (開迴路)中，依講義完成電路接線後
- (1)將Vis之電源移除後兩端再短路，此電路是否依然可以執行模擬？ 是 。
- (2)出現何種訊息？ Syntax error. 。
- (3)移除Vis電源再短路兩端點，若不變動 Ipri 電流探棒在電路圖上的位置(沒有產生連接的紅點)，欲維持原先各點的輸出波形，應如何修改電路？ 將電流探棒接到下方Vd 。
- 請貼上此修改後之電路圖於下方。

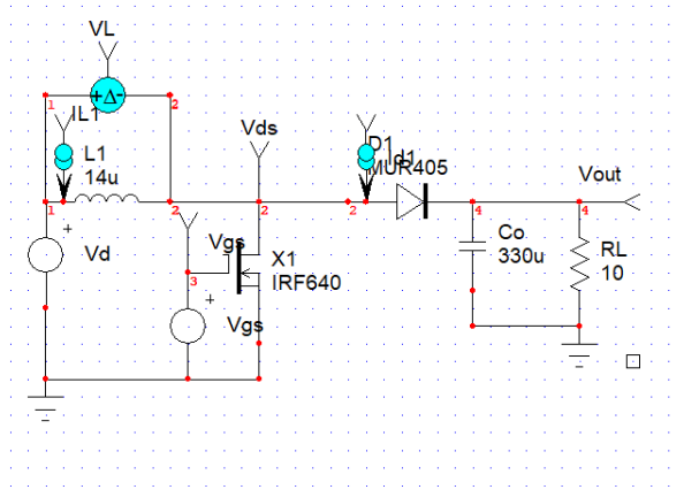


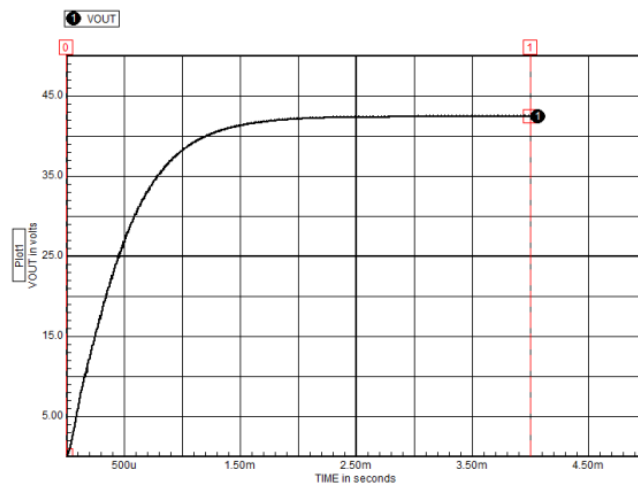
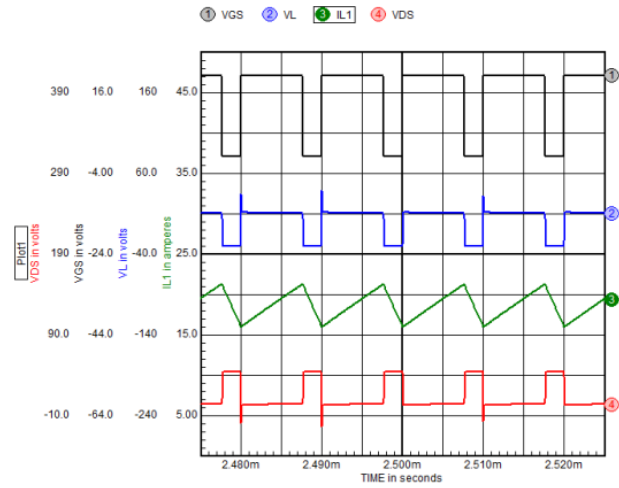
- 3-2. 實習三 直流至直流轉換電路--升壓型轉換器 Step-Up Converter (開迴路)中，貼上此電路之Duty 為25%及75%時的 (1)電路圖與 (2)波形圖於下方。(請參考實習三升壓型轉換器的波形圖，包含CH1:開極驅動電壓Vgs、CH2:電感電壓VL、CH3:電感電流IL1、CH4:MOS汲源極電壓Vds、CH5:輸出電壓Vout等一起顯示的波形圖，波形顯示週期為 5us/div)
- Duty 25%：





Duty: 75%

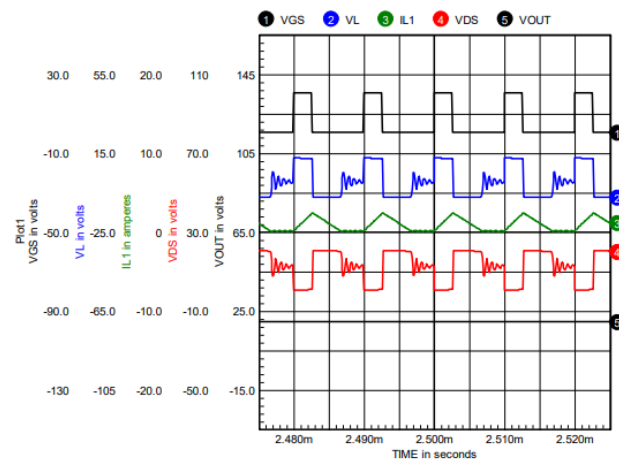
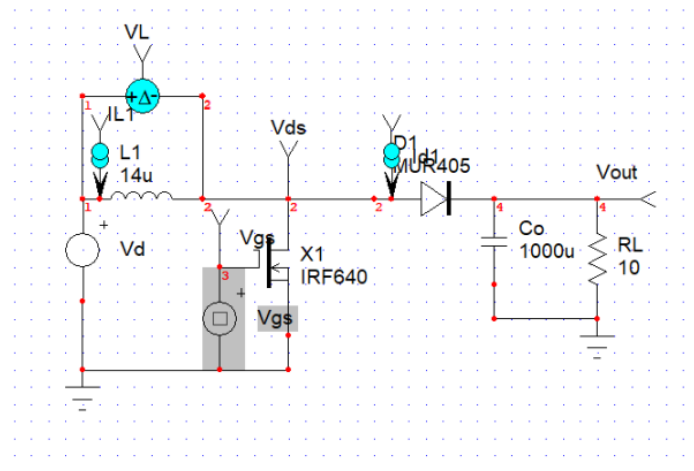




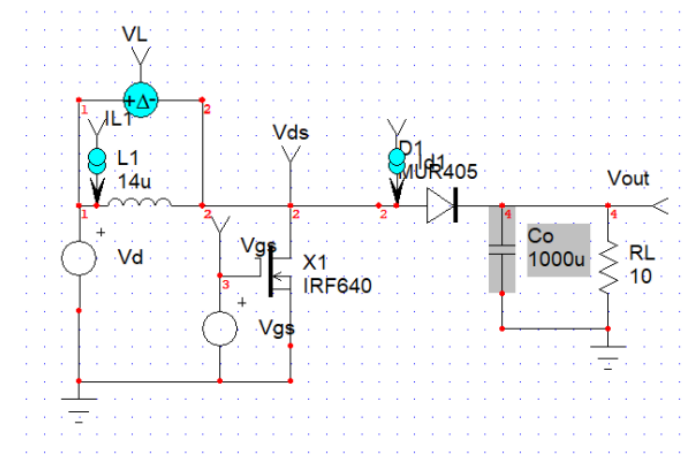
3-3. 若將實習三 直流至直流轉換電路-升壓型轉換器 Step-Up Converter (開迴路)中之電容 C_o 由 $330\mu\text{F}$ 改為 $1000\mu\text{F}$ ，請重作練習 3-2，並且將(1)電路圖與 (2)波形圖貼

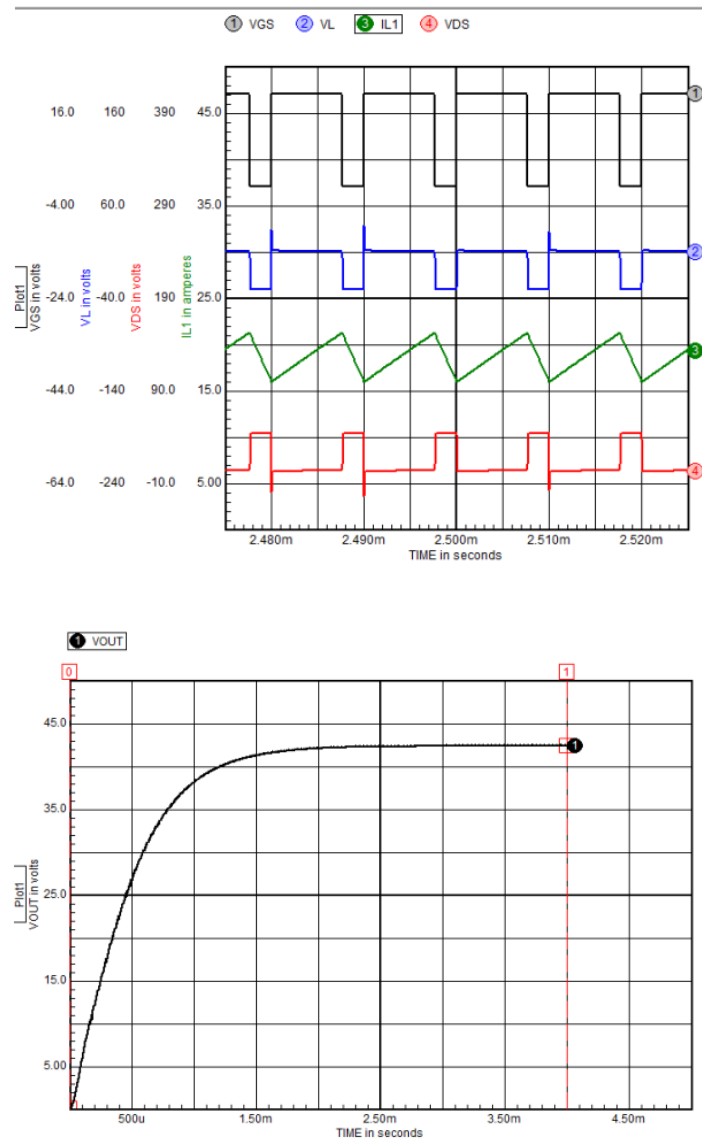
於下方。

Duty: 25%



Duty: 75%





二、實習報告內容

1. 摘要說明

- **降壓型轉換器**：又稱 Buck Converter，主要功能是將較高的非穩壓直流輸入電壓 (V_g) 轉換為較低、受控且穩定的直流輸出 (V_{out})。

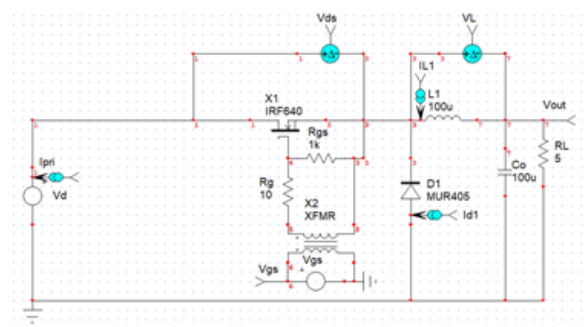
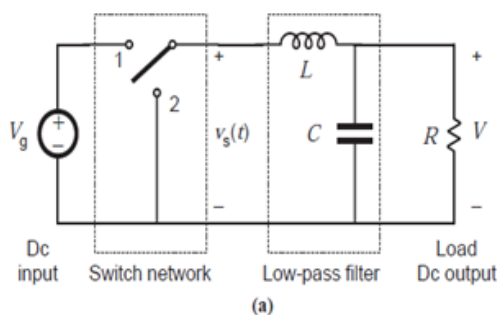
應用於需要將固定高直流電壓降至可變或固定低直流電壓的情形。

- **高效能與應用：** 高效率是降壓型轉換器設計目標。透過開關模式 (Switched-Mode) 操作將功率損耗最小化，通常可達到 70%~95% 轉換效率。

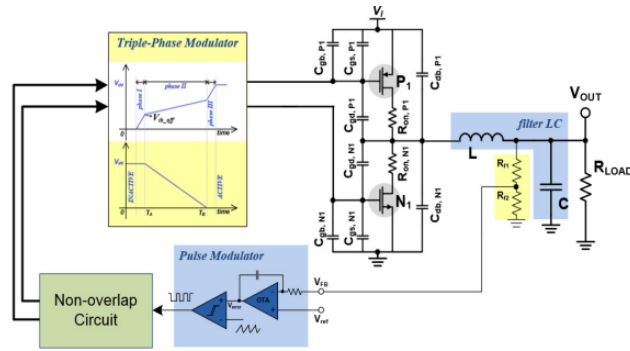
低損耗可減少熱量產生(簡化散熱設計並降低系統成本)，使其成為各種電子設備中不可或缺的電源模組(個人電腦、筆記型電腦、智慧型手機處理器供電，以及電信設備、太陽能板最大功率點追蹤 (MPPT) 系統和電動車充電器等)。

文獻提出改良型的 Buck Converter，主要目的是為了減少 MOSFET 快速切換時所產生的供應電壓震盪 (supply bouncing)。透過引入 slew-rate modulator (斜率速率調變器) 來延長 PWM 輸出訊號的上升/下降時間，取代傳統的輸出級電路以減輕 RC 震盪。

2. 電路架構



傳統 Buck converter



加入 slew-rate modulator 的 Buck converter

以實習模擬參數為例：

○ 直流輸入電壓 (V_g)：

- $V_d = 24V$
- 作用：未經穩壓的原始直流電壓源，範圍和穩定性因應用而異。

○ 開關網路 (Switch network)：

- 作用：核心功能單元，由一個或多個功率半導體開關（如 MOSFETs、IGBTs）以及二極體、電感等無源儲能元件構成。其主要功能是週期性地斷開 V_g 與負載之間的連接(產生脈衝電壓 vs (t))。
- 原理：透過高速切換[導通 (ON) 和截止 (OFF)]兩種狀態之間快速轉換，理想開關導通時呈低阻抗（近似短路），截止時呈高阻抗狀態（近似開路），減少導通損耗和截止損耗，實現高效率能量傳輸。

在脈衝寬度調變 (PWM)控制下，週期性地導通和截止，將輸入

直流電壓 V_g 「斬波」成脈衝電壓 $v_s(t)$ 。

- 以實習的 PWM 驅動訊號 (V_{gs}) 為例：
 - 脈衝寬度 (Pulse width, t_{on}) : $2.1\mu s$ 。
 - 週期 (Period, T_s) : $10\mu s$ 。
 - 開關頻率 (f_s) = $1/T_s = 1/(10\mu s) = 100kHz$ 。
 - 佔空比 (D) = $t_{on} / T_s = 2.1\mu s / 10\mu s = 0.21$ 。

○ 低通濾波器 (Low-pass filter) :

- 作用： 通常由電感 (L) 和電容 (C) 組成。濾除開關輸出的脈衝電壓 $v_s(t)$ 高頻諧波，保留直流平均值，產生平滑穩定的直流輸出電壓 V 。
- 原理： 負載需要一個平穩的直流電壓。LC 濾波器利用電感和電容儲能特性，對脈衝電壓進行時間平均化，實現高頻雜訊濾除和直流成分提取。
 - 電感 (L)： 穩流，當開關導通時，電感電流線性增加，將能量儲存於磁場中；當開關截止時，電感產生反向電壓維持電流流動，將儲存的能量釋放到負載和電容中。
 - 電容 (C)： 穩壓，電容並聯於負載，吸收電感的脈衝電流並在開關斷開期間提供瞬時能量，穩定輸出電壓減少漣波。

- Slew-Rate Modulator :

在實際電路當 MOSFET 切換速度過快，會造成寄生電感與電容形成的電磁干擾（EMI）與瞬間震盪，尤其在高密度整合電路（SoC）中會干擾其他模組的正常運作。(Supply Bouncing)

Slew-Rate Modulator（SRM）可用來控制 PWM 訊號斜率，使原本方波的上升與下降變得更緩慢，類似鋸齒狀或三角波。

過程類似在 PWM 與功率開關間加上「阻尼器」，降低開關瞬間電壓變化（ dV/dt ），減少由寄生元件引起的震盪與 EMI。

3. 電路的原理

- 開關模式操作（Switched-Mode Operation）：

透過高頻率的開關動作結合儲能元件（電感與電容）特性，對輸入直流電壓進行「斬波」和「濾波」實現高效的電壓轉換和穩壓功能。(透過控制功率開關的佔空比 (D) 來調節輸出電壓為核心原理)

- 一般 Step Down Converter：

- Switch ON ($0 < t \leq D \cdot T_s$)：

當 V_{gs} 訊號為高電位時 MOSFET 導通，D1 由於處於反向截止狀態，此時 V_g 直接施加在 L1 和輸出濾波器（ C_o 和 R_L ）的串聯迴路上。

V_g 向 $L1$ 供給能量，使電感電流 (I_{L1}) 線性增加。電感的電壓為 $V_g - V_{out}$ ，電感將電能轉化為磁能儲存起來。同時部分能量直接通過濾波器供給 R_L 並為 C_o 充電。

○ Switch OFF ($D \cdot T_s < t \leq T_s$) :

V_{gs} 訊號為低電位時 MOSFET 開路，此時 $L1$ 會產生感應電壓使電流透過續流 $D1$ 繼續流動，形成 $L1 - D1 - C_o - R_L$ 的閉合迴路。

$L1$ 釋放儲存的磁能轉化為電能，透過 $D1$ 繼續向 R_L 供電，同時為 C_o 充電或補充其向負載放電的能量，電感電流 (I_{L1}) 線性減小。此時電感的電壓為 $-V_{out}$ 。

○ 伏秒平衡：

轉換器達到穩態運行時，電感在一個開關週期內的平均電壓必須為零。

導通期間電感電壓面積（伏秒積）： $A_{on} = (V_g - V_{out}) \cdot t_{on} = (V_g - V_{out}) \cdot D \cdot T_s$

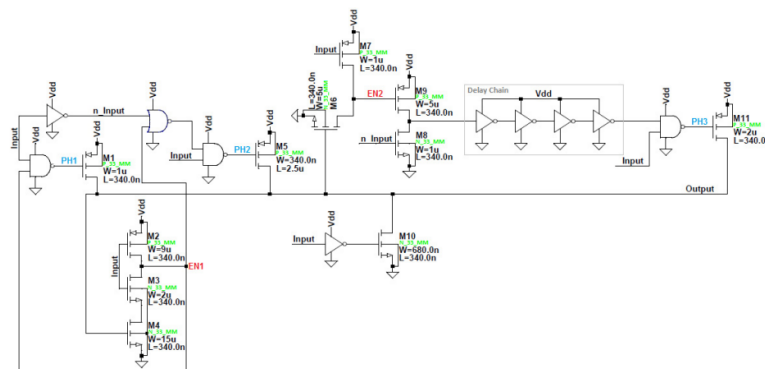
截止期間電感電壓面積（伏秒積）： $A_{off} = (-V_{out}) \cdot t_{off} = (-V_{out}) \cdot (1-D) \cdot T_s$

$$A_{on} + A_{off} = 0$$

簡化後可得： $V_g D - V_{out} = 0 \gg V_{out} = D \cdot V_g$

V_{out} 始終小於或等於輸入電壓 V_g ($0 \leq D \leq 1$)，透過調整佔空比 D 控制輸出電壓。

○ Slew-Rate Modulator：



SRM 使用三顆 MOSFET (M1、M5、M11) 並聯於訊號路徑，形成三段電流控制通路，每一顆 PMOS 是獨立導通、各自控制一部分電流量，每次導通一顆，總電流逐步增加達成控制斜率的目的。

Phase	開啟的 PMOS	輸出電流大小	輸出斜率 (dV/dt)	說明
PHASE1	M1	中間	緩慢上升	初始輸出緩慢 充電
PHASE2	M5	最小	更緩	延長主要斜率

				調整階段
PHASE3	M11	最大	快速完成上升	快速導通結尾，減少延遲

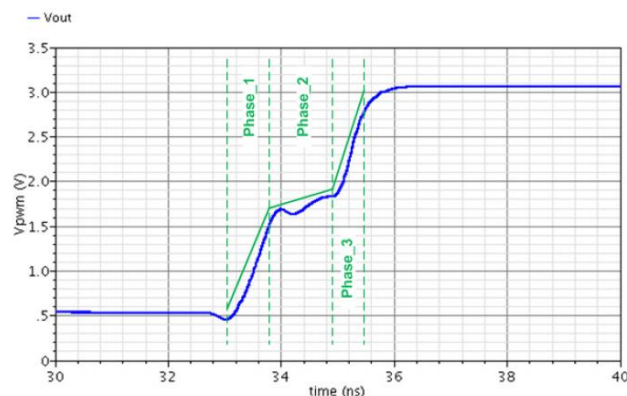


Figure 2. Transient analysis detailed waveform of output signal of the slew-rate control output stage.

4. 電路的功能或波形量測

- 損耗分析：

文獻分析降壓轉換器功率級的寄生阻抗，並著重功耗損耗，包括導通損耗（conduction losses）和開關損耗（switching losses）。

- 模擬結果與電路配置：

研究中使用的降壓轉換器採用傳統同步降壓配置，實現電壓模式 PWM 控制與主極點補償，並透過三相轉換速率調變器調變訊號以降低電源反彈，隨後應用於 MOSFET。

轉換器為 UMC 0.18um CMOS 製程設計，預期將 3.3V 轉換為 1.8V，

提供 30mA 輸出電流。

輸出電壓漣波 (ΔV_{out}) 需小於 1%，電流漣波 (ΔI_L) 低於 50%，開關頻率為 100MHz。

相關算式：

$$L_{\min} = \frac{V_{out}}{\Delta i_L \cdot f_s} (1 - D)$$

$$\Delta V_{out} = \frac{V_{out} \cdot (1 - D)}{8 \cdot L \cdot C \cdot f_s^2} \quad \eta = 100 \times \frac{P_{out}}{P_{out} + P_{buck}}$$

- 結果：

- 輸出電壓暫態分析：輸出電壓漣波 (Ripple)

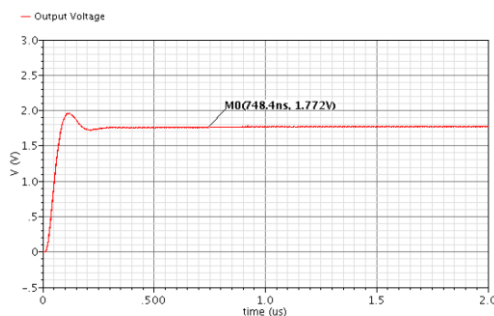
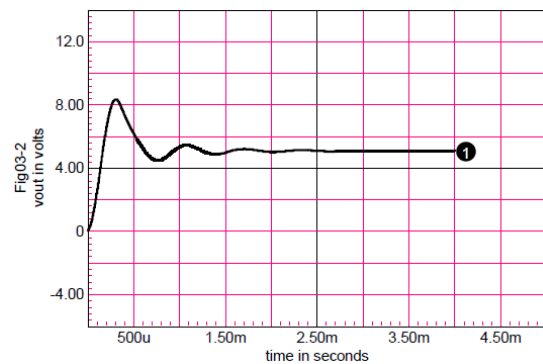


Figure 7. Output voltage transient analysis of the buck converter.



左圖為文獻結果，輸出電壓約在 300ns 時穩定於 1.8V

右圖則是實習電路，從波形分析可以發現加上 SRM 做控制回授明顯減少了 RC Damping，並且文獻花了約僅右圖電路的 0.01% 時間就達到了平衡(輸入電壓差也有影響)，明顯變得穩定。

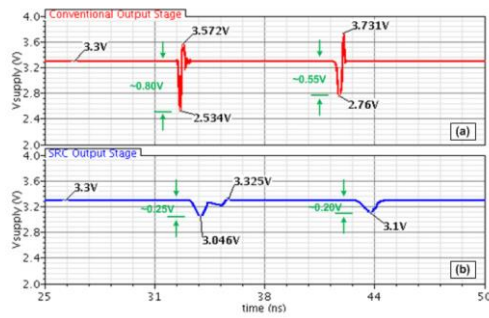


Figure 10. Transient analysis detailed waveforms of VDD with (a) the conventional output stage, and (b) the slew-rate control output stage, respectively.

最後文章中也有將放上 SRM 前後做對比，傳統輸出震盪達 800mV，使用 slew-rate 控制器後降至 250mV，也說明了此設計相較傳統更適合 SoC 應用。

○ 輸出漣波與電感電流漣波（Ripple）

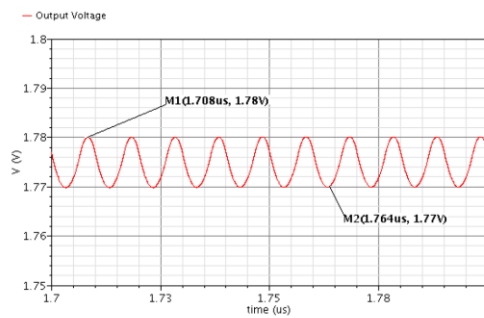


Figure 8. Output voltage ripple transient analysis of the buck converter.

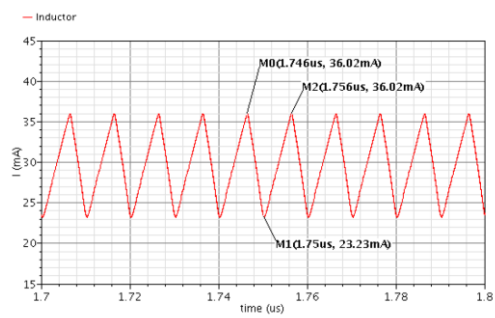
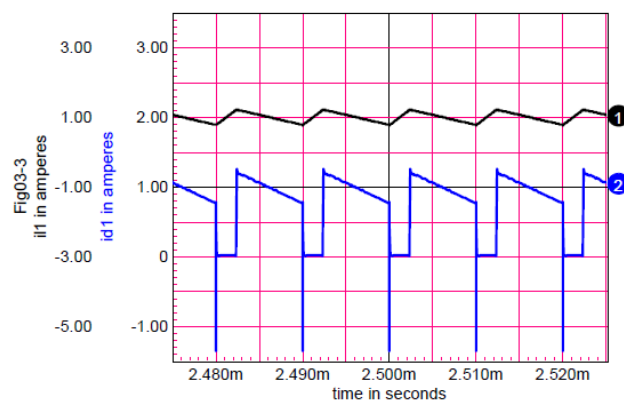


Figure 9. Inductor's peak-to-peak current ripple (ΔI_L) transient analysis of the buck converter.

Fig03-3 ❶ il1 ❷ id1



- 輸出電壓的模擬結果漣波為 10mV，為 1% of 1.8V(預期)，峰對峰電流約為 30mA(<50%輸出電流)

電流漣波變化相較實習也更小。

5. 電路可應用的範圍

- 電子產品：

- 行動裝置內部用於電池組，供應給處理器、記憶體、顯示螢幕背光和其他低壓數位晶片的精確直流電壓。(System on Chip (系統單晶片))

- 家用電器：LED 照明驅動電路、智慧家居設備內部電源管理，提供穩定且符合模組的電壓。

- 工業控制與自動化：

- 工業控制器 (PLCs)：將標準工業電源電壓降壓轉換為控制器內部邏輯電路所需的較低電壓。

- 感測器與驅動器：為工業感測器、馬達驅動控制電路以及自動化設備提供精確穩定的電源。

- 汽車電子：

- 車載電源管理：電動車的高壓電池組轉換為車載資訊娛樂系統、車輛電子控制單元 (ECU) 和其他電子模組所需的低壓電源。

- 電動車 (EV) / 混合動力車 (HEV)：在電池管理系統 (BMS) 中，將高壓電池組的電壓有效降壓，為輔助低壓系統供電。

- 再生能源系統：

- 太陽能光伏系統：與最大功率點追蹤 (MPPT) 控制器結合，將太陽能板輸出的可變電壓轉換為適合電池充電或直接連接到電網的穩定直流電壓。

- 電信與資料中心：

- 伺服器電源單元 (PSU)：將數據中心的高電壓直流母線高效降壓轉換為伺服器處理器和記憶體所需的極低電壓，最小化熱量產生並提升效率。
- 電信基站：為通信設備的各級電路（射頻模組、數位信號處理器）提供精確穩定的電源。

6. 電路的出處來源及其他相關電路

- 模擬與電路圖來源：

- 本實習課程使用 Ispice 軟體製作的模擬圖。
- 參考資料的論文中。

- 其他相關電路：

- 升壓型轉換器 (Boost Converter)：

一種 DC-DC 轉換器，用來將輸入電壓提升為比原本更高的輸

出電壓。其基本結構包含：

- 一個功率開關（MOSFET）
- 一個電感器
- 一個二極體
- 一個輸出濾波電容

運作原理：

- 開關導通：電感儲存能量，輸出由電容供電。
- 開關關閉：電感釋放能量，與輸入共同供電至負載，電壓因此被“推升”。

這種電路效率高、結構簡單，常見於太陽能供電、LED 驅動、鋰電池升壓等應用。

○ 降升壓型轉換器（Buck-Boost Converter）：

是種能「提升」也能「降低」輸出電壓的 DC-DC 轉換器。它可以根據負載或輸入條件動態調整輸出電壓，讓系統更穩定、更有彈性。常見的電路架構有兩種：

- 非同步型 Buck-Boost（基本型）
- 同步四開關 Buck-Boost（高效能版）

運作邏輯：

- 輸入電壓低時：作為升壓器用。

- 輸入電壓高時：作為降壓器用。

這種電路廣泛應用於電池供電設備（如行動電源）、電動車電源模組、工業裝置等，特別適合輸入電壓不穩的場景。

三、課程心得與討論

王奕傑： 這次的降、升壓型轉換器實作與模擬讓我們對電源電路的運作原理有更深入的理解，上學期課程只有聽老師講解，實際跑電路模擬的時候，也學到不少細節：像是 PWM 訊號怎麼調會影響輸出電壓、選錯二極體型號會讓波形亂跳等等，這些都是實作過就忘不了的經驗。另外，也更體會到設計一個穩定的 DC-DC 轉換器不只是“畫圖+接線”那麼簡單，還要顧慮效率、元件選擇、控制邏輯等等。

這堂課讓我學到的不只是電路知識，還有解決問題與 debug 的能力，雖然過程有點燒腦，但真的蠻值得的。

張博翔： 此學期的課程幫助我對於上學期知識整合與運用有更進一步理解，並且因為實作過程中總會遇到一些問題，如 debug、波型改變原因等等，可以重新審視自己那些地方不熟練以及需要查找哪些方向的資料，如中央抽頭式電路、當時搞錯的負載意思等等，都是基本卻非常重要的觀念，另外也更加了解電力電子該分析什麼以及對哪些輸出感興趣。

也非常感謝老師這個學期的耐心指導與解惑，這一年的學習讓我對於電機系有更加認識並也補起了很多基礎，感激不盡！老師辛苦了！

四、組員分工內容與工作份量比例

組員姓名	負責內容	工作比例
張博翔	1、2、3、4 部分	50%
王奕傑	4、5、6 部分	50%

五、註記的兩篇主題資料全文

Article in Wiley Encyclopedia of Electrical and Electronics Engineering

DC-DC Power Converters

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Dc-dc power converters are employed in a variety of applications, including power supplies for personal computers, office equipment, spacecraft power systems, laptop computers, and telecommunications equipment, as well as dc motor drives. The input to a dc-dc converter is an unregulated dc voltage V_g . The converter produces a regulated output voltage V , having a magnitude (and possibly polarity) that differs from V_g . For example, in a computer off-line power supply, the 120 V or 240 V ac utility voltage is rectified, producing a dc voltage of approximately 170 V or 340 V, respectively. A dc-dc converter then reduces the voltage to the regulated 5 V or 3.3 V required by the processor ICs.

High efficiency is invariably required, since cooling of inefficient power converters is difficult and expensive. The ideal dc-dc converter exhibits 100% efficiency; in practice, efficiencies of 70% to 95% are typically obtained. This is achieved using switched-mode, or chopper,

circuits whose elements dissipate negligible power. Pulse-width modulation (PWM) allows control and regulation of the total output voltage. This approach is also employed in applications involving alternating current, including high-efficiency dc-ac power converters (inverters and power amplifiers), ac-ac power converters, and some ac-dc power converters (low-harmonic rectifiers).

A basic dc-dc converter circuit known as the *buck converter* is illustrated in Fig. 1. A single-pole double-throw (SPDT) switch is connected to the dc input voltage V_g as shown. The switch output voltage

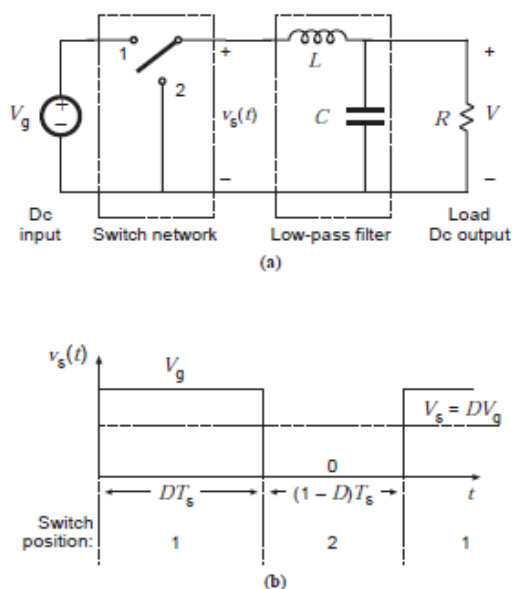


Figure 1. The buck converter consists of a switch network that reduces the dc component of voltage, and a low-pass filter that removes the high-frequency switching harmonics: (a) schematic, (b) switch voltage waveform.

$v_s(t)$ is equal to V_g when the switch is in position 1, and is equal to zero when the switch is in position 2. The switch position varies periodically, such that $v_s(t)$ is a rectangular waveform having period T_g and duty cycle D . The duty cycle is equal to the fraction of time that the switch is connected in position 1, and hence $0 \leq D \leq 1$. The switching frequency f_g is equal to $1/T_g$. In practice, the SPDT switch is realized using semiconductor devices such as diodes, power MOSFETs, IGBTs, BJTs, or thyristors. Typical switching frequencies lie in the range 1 kHz to 1 MHz, depending on the speed of the semiconductor devices.

The switch network changes the dc component of the voltage. By Fourier analysis, the dc component of a waveform is given by its average value. The average value of $v_s(t)$ is given by

$$V_s = \frac{1}{T_g} \int_0^{T_g} v_s(t) dt = DV_g \quad (1)$$

The integral is equal to the area under the waveform, or the height V_g multiplied by the time DT_g . It can be seen that the switch network reduces the dc component of the voltage by a factor equal to the duty cycle D . Since $0 \leq D \leq 1$, the dc component of V_s is less than or equal to V_g .

The power dissipated by the switch network is ideally equal to zero. When the switch contacts are closed, then the voltage across the contacts is equal to zero and hence the power dissipation is zero. When the switch contacts are open, then there is zero current and the power dissipation is again equal to zero. Therefore, the ideal switch network is able to change the dc component of voltage without dissipation of power.

In addition to the desired dc voltage component V_s , the switch waveform $v_s(t)$ also contains undesired harmonics of the switching frequency. In most applications, these harmonics must be removed, such that the converter output voltage $v(t)$ is essentially equal to the dc component $V = V_s$. A low-pass filter is employed for this purpose. The converter of Fig. 1 contains a single-section L - C low-pass filter. The filter has corner frequency f_0 given by

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad (2)$$

The corner frequency f_0 is chosen to be sufficiently less than the switching frequency f_g , so that the filter essentially passes only the dc component of $v_s(t)$. To the extent that the inductor and capacitor are ideal, the filter removes the switching harmonics without dissipation of power. Thus, the converter produces a dc output voltage whose magnitude is controllable via the duty cycle D , using circuit elements that (ideally) do not dissipate power.

The conversion ratio $M(D)$ is defined as the ratio of the dc output voltage V to the dc input voltage V_g under steady-state conditions:

$$M(D) = \frac{V}{V_g} \quad (3)$$

For the buck converter, $M(D)$ is given by

$$M(D) = D \quad (4)$$

This equation is plotted in Fig. 2. It can be seen that the dc output voltage V is controllable between 0 and V_g by adjustment of the duty cycle D .

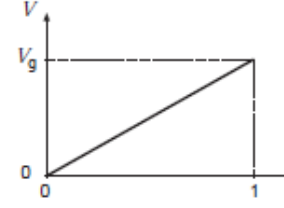


Figure 2. Buck converter dc output voltage V vs. duty cycle D .

Figure 3 illustrates one way to realize the switch network in the buck converter, using a power MOSFET and diode. A gate drive circuit switches the MOSFET between the conducting (on) and blocking (off) states, as commanded by a logic signal $\delta(t)$. When $\delta(t)$ is high (for $0 < t < DT_s$), then MOSFET Q_1 conducts with negligible drain-to-source voltage. Hence, $v_s(t)$ is approximately equal to V_g and the diode is reverse-biased. The positive inductor current $i_L(t)$ flows through the MOSFET. At time $t = DT_s$, $\delta(t)$ becomes low, commanding MOSFET Q_1 to turn off. The inductor current must continue to flow; hence, $i_L(t)$ forward-biases diode D_1 , and $v_s(t)$ is now approximately equal to zero. Provided that the inductor current $i_L(t)$ remains positive, then diode D_1 conducts for the remainder of the switching period. Diodes that operate in the manner are called *freewheeling diodes*.

Since the converter output voltage $v(t)$ is a function of the switch duty cycle D , a control system can be constructed that varies the duty cycle to cause the output voltage to follow a given reference v_r . Figure 3 illustrates the block diagram of a simple converter feedback system. The output voltage is sensed using a voltage divider, and is compared with an accurate dc reference voltage v_r . The resulting error signal is passed through an op-amp compensation

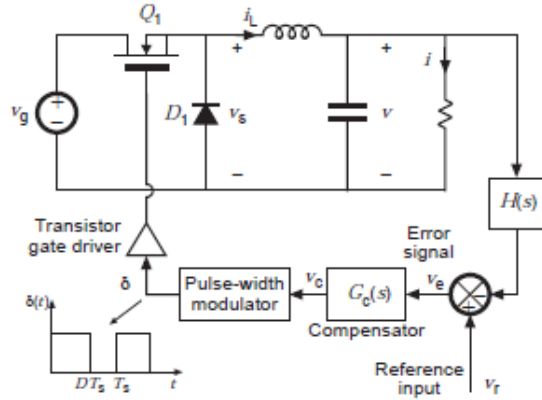


Figure 3. Realization of the ideal SPDT switch using a transistor and freewheeling diode. In addition, a feedback loop is added for regulation of the output voltage.

network. The analog voltage $v_o(t)$ is next fed into a *pulse-width modulator*. The modulator produces a switched voltage waveform that controls the gate of the power MOSFET Q_1 . The duty cycle D of this waveform is proportional to the control voltage $v_o(t)$. If this control system is well designed, then the duty cycle is automatically adjusted such that the converter output voltage v follows the reference voltage v_r , and is essentially independent of variations in v_g or load current.

Converter circuit topologies

A large number of dc-dc converter circuits are known that can increase or decrease the magnitude of the dc voltage and/or invert its polarity [1-5]. Figure 4 illustrates several commonly used dc-dc converter circuits, along with their respective conversion ratios. In each example, the switch is realized using a power MOSFET and diode; however, other semiconductor switches such as IGBTs, BJTs, or thyristors can be substituted if desired.

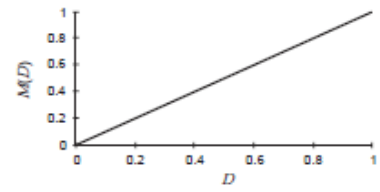
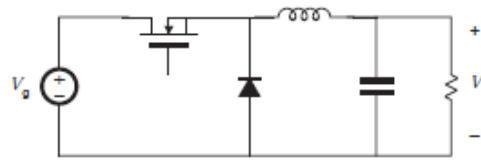
The first converter is the buck converter, which reduces the dc voltage and has conversion ratio $M(D) = D$. In a similar topology known as the boost converter, the positions of the switch and inductor are interchanged. This converter produces an output voltage V that is greater in magnitude than the input voltage V_g . Its conversion ratio is $M(D) = 1/(1 - D)$.

In the buck-boost converter, the switch alternately connects the inductor across the power input and output voltages. This converter inverts the polarity of the voltage, and can either increase or decrease the voltage magnitude. The conversion ratio is $M(D) = -D/(1 - D)$.

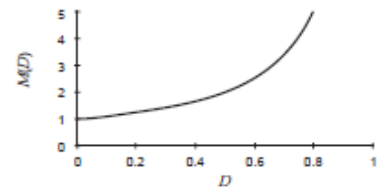
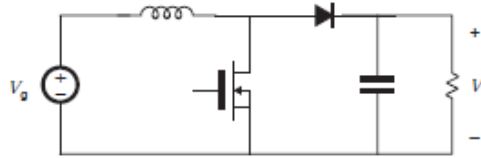
The Cuk converter contains inductors in series with the converter input and output ports. The switch network alternately connects a capacitor to the input and output inductors. The conversion ratio $M(D)$ is identical to that of the buck-boost converter. Hence, this converter also inverts the voltage polarity, while either increasing or decreasing the voltage magnitude.

The single-ended primary inductance converter (SEPIC) can also either increase or decrease the voltage magnitude. However, it does not invert the polarity. The conversion ratio is $M(D) = D/(1 - D)$.

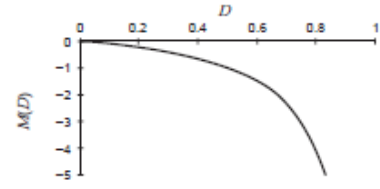
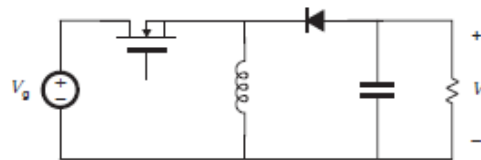
Buck converter



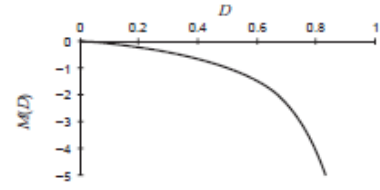
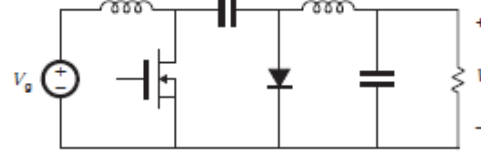
Boost converter



Buck-boost converter



Cuk converter



SEPIC

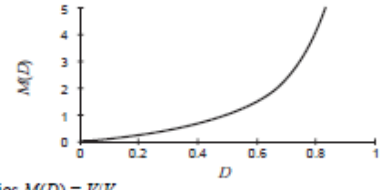
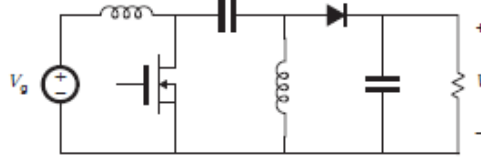


Figure 4. Several basic dc-dc converters and their dc conversion ratios $M(D) = V/V_g$.

Analysis of converter waveforms

Under steady-state conditions, the voltage and current waveforms of a dc-dc converter can be found by use of two basic circuit analysis principles. The principle of *inductor volt-second balance* states that the average value, or dc component, of voltage applied across an ideal inductor winding must be zero. This principle also applies to each winding of a transformer or other multiple winding magnetic devices. Its dual, the principle of *capacitor amp-second or charge balance*, states that the average current that flows through an ideal capacitor must be zero. Hence, to determine the voltages and currents of dc-dc converters operating in periodic steady state, one averages the inductor current and capacitor voltage waveforms over one switching period, and equates the results to zero.

The equations are greatly simplified by use of a third artifice, the *small ripple approximation*. The inductor currents and capacitor voltages contain dc components, plus *switching ripple* at the switching frequency and its harmonics. In most well designed converters, the switching ripple is small in magnitude compared to the dc components. For inductor currents, a typical value of switching ripple at maximum load is 10% to 20% of the dc component of current. For an output capacitor voltage, the switching ripple is typically required to be much less than 1% of the dc output voltage. In both cases, the ripple magnitude is small compared with the dc component, and can be ignored.

As an example, consider the boost converter of Fig. 5(a). A resistor R_L is included in series with the inductor, to model the resistance of the inductor winding. It is desired to determine simple expressions for the output voltage V , inductor current I_L , and efficiency η . Typical inductor voltage and capacitor current waveforms are sketched in Fig. 5(b).

With the switch in position 1, the inductor voltage is equal to $v_L(t) = V_g - i_L(t)R_L$. By use of the small ripple approximation, we can replace $i_L(t)$ with its dc component I_L , and hence obtain $v_L(t) \approx V_g - I_L R_L$. Likewise, the capacitor current is equal to $i_C(t) = -v(t)/R$, which can be approximated as $i_C(t) \approx -V/R$.

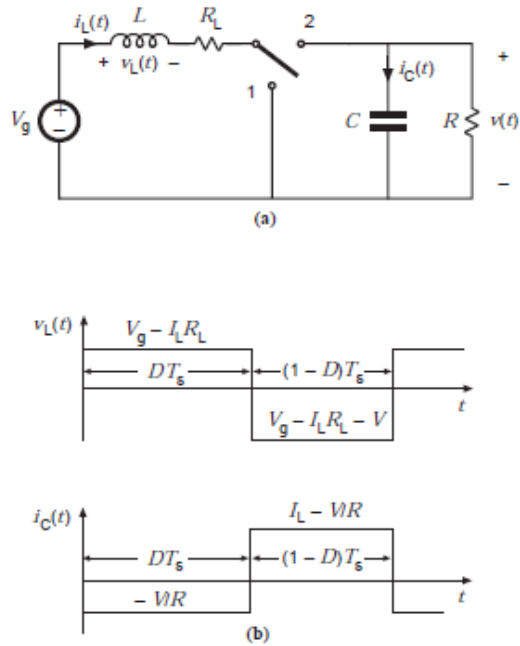


Figure 5. A nonideal boost converter: (a) schematic, (b) inductor voltage and capacitor current waveforms.

When the switch is in position 2, the inductor is connected between the input and output voltages. The inductor voltage can now be written $v_L(t) = V_g - i_L(t)R_L - v(t) \approx V_g - I_L R_L - V$. The capacitor current can be expressed as $i_C(t) = i_L(t) - v(t)/R \approx I_L - V/R$.

When the converter operates in steady state, the average value, or dc component, of the inductor voltage waveform $v_L(t)$ must be equal to zero. Upon equating the average value of the $v_L(t)$ waveform of Fig. 5(b) to zero, we obtain

$$0 = D(V_g - I_L R_L) + (1 - D)(V_g - I_L R_L - V) \quad (5)$$

Likewise, application of the principle of capacitor charge balance to the capacitor current waveform of Fig. 5(b) leads to

$$0 = D\left(-\frac{V}{R}\right) + (1 - D)\left(I - \frac{V}{R}\right) \quad (6)$$

Equations (5) and (6) can now be solved for the unknowns V and I_L . The result is

$$\frac{V}{V_g} = \frac{1}{(1 - D)} \frac{1}{\left(1 + \frac{R_L}{(1 - D)^2 R}\right)} \quad (7)$$

$$I_L = \frac{V_g}{(1 - D)^2 R} \frac{1}{\left(1 + \frac{R_L}{(1 - D)^2 R}\right)} \quad (8)$$

Equation (7) is plotted in Fig. 6, for several values of R_L/R . In the ideal case when $R_L = 0$, the voltage conversion ratio $M(D)$ is equal to one at $D = 0$, and tends to infinity as D approaches one. In the practical

case where some small inductor resistance R_L is present, the output voltage tends to zero at $D = 1$. In addition, it can be seen that the inductor winding resistance R_L (and other loss elements as well) limits the maximum output voltage that the converter can produce. Obtaining a given large value of V/V_g requires that the winding resistance R_L be sufficiently small.

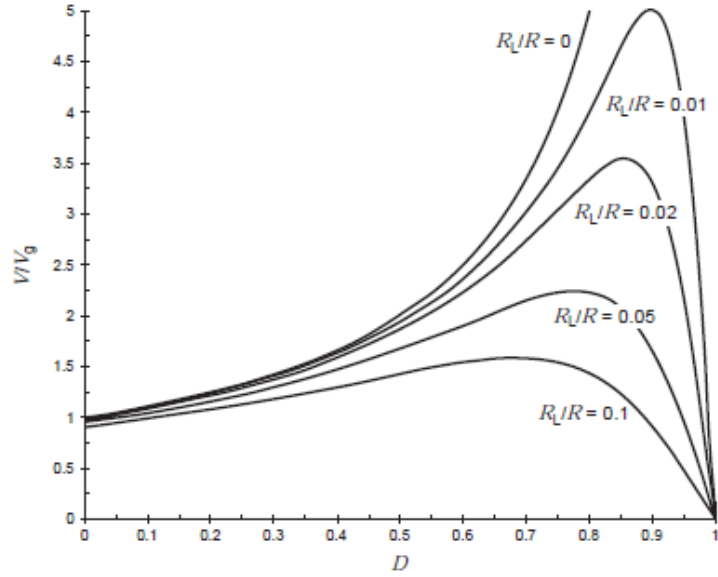


Figure 6. Output voltage vs. duty cycle, for the nonideal boost converter of Fig. 5.

The converter efficiency can also be determined. For this boost converter example, the efficiency is equal to

$$\eta = \frac{P_{out}}{P_{in}} = \frac{(V^2/R)}{(V_b I_L)} \quad (9)$$

Substitution of Eqs. (7) and (8) into Eq. (9) leads to

$$\eta = \frac{1}{1 + \frac{R_L}{(1-D)^2 R}} \quad (10)$$

This expression is plotted in Fig. 7, again for several values of R_L/R . It can be seen that, to obtain high efficiency, the inductor winding resistance R_L should be much smaller than $(1-D)^2 R$. This is much easier to accomplish at low duty cycles, where $(1-D)$ is close to unity, that at high duty cycles where

$(1-D)$ approaches zero. Consequently, the efficiency is high at low duty cycles, but decreases rapidly to zero near $D = 1$. This behavior is typical of converters having boost or buck-boost characteristics.

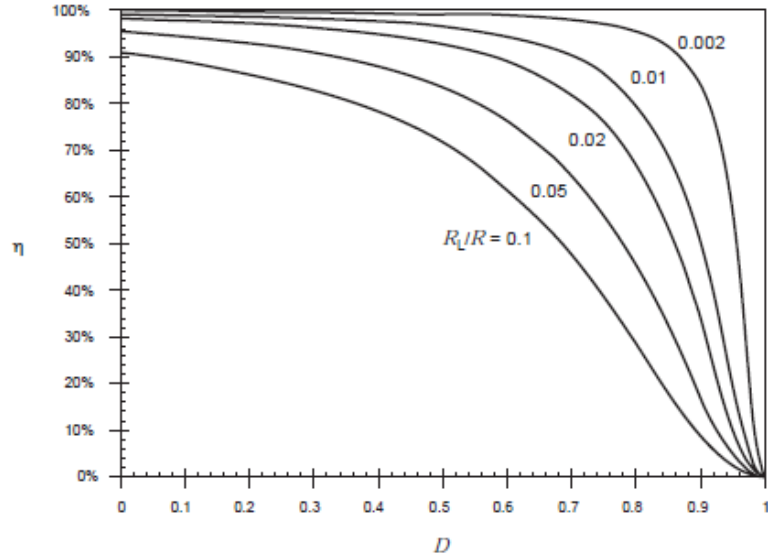


Figure 7. Efficiency vs. duty cycle, for the nonideal boost converter of Fig. 5.

Transformer isolation

In the majority of applications, it is desired to incorporate a transformer into the switching converter, to obtain dc isolation between the converter input and output. For example, in off-line power supply applications, isolation is usually required by regulatory agencies. This isolation could be obtained by simply connecting a 50 Hz or 60 Hz transformer at the power supply ac input terminals. However, since transformer size and weight vary inversely with frequency, incorporation of the transformer into the converter can make significant improvements: the transformer then operates at the converter switching frequency of tens or hundreds of kilohertz. The size of modern ferrite power transformers is minimized at

operating frequencies ranging from several hundred kilohertz to roughly one Megahertz. These high frequencies lead to dramatic reductions in transformer size.

When a large step-up or step-down conversion ratio is required, the use of a transformer can allow better converter optimization. By proper choice of the transformer turns ratio, the voltage or current stresses imposed on the transistors and diodes can be minimized, leading to improved efficiency and lower cost.

Multiple dc outputs can also be obtained in an inexpensive manner, by adding multiple secondary windings and converter secondary-side circuits. The secondary turns ratios are chosen to obtain the desired output voltages. Usually, only one output voltage can be regulated, via control of the converter duty cycle, so wider tolerances must be allowed for the auxiliary output voltages. *Cross regulation* is a measure of the variation in an auxiliary output voltage, given that the main output voltage is regulated perfectly.

The basic operation of transformers in most power converters can be understood by replacing the transformer with the simplified model illustrated in Fig. 8. The model neglects losses and imperfect coupling between windings; such phenomena are usually considered to be converter nonidealities. The model consists of an ideal transformer plus a shunt inductor known as the magnetizing inductance L_M . This inductor models the magnetization of the physical transformer core, and hence it must obey all of the usual rules for inductors. In particular, volt-second balance must be maintained on the magnetizing inductance. Furthermore, since the voltages of all windings of the ideal transformer are proportional, volt-second balance must be maintained for each winding. Failure to achieve volt-second balance leads to transformer saturation and, usually, destruction of the converter. The means by which transformer volt-second balance is achieved is known as the *transformer reset* mechanism.

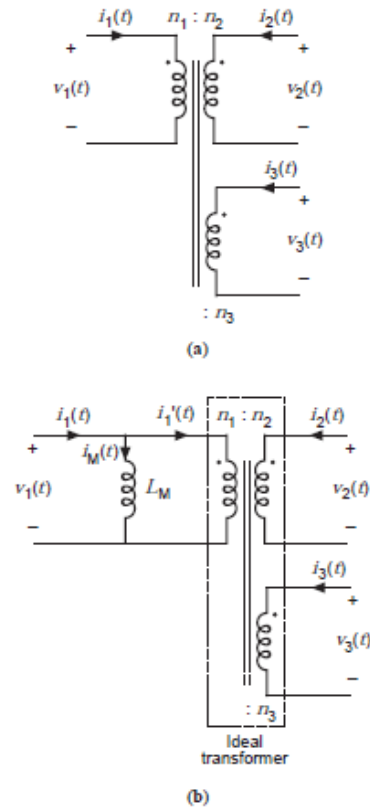


Figure 8. Modeling a physical transformer such that its basic operation within an isolated dc-dc converter can be understood: (a) transformer schematic symbol, (b) equivalent circuit model that includes magnetizing inductance L_M and an ideal transformer.

There are several ways of incorporating transformer isolation into any dc-dc converter. The full-bridge, half-bridge, forward, and push-pull converters are commonly used isolated versions of the buck converter. Similar isolated variants of the boost converter are known. The flyback converter is an isolated version of the buck-boost converter. Isolated variants of the SEPIC and Cuk converter are also known. The full-bridge, forward, and flyback converters are briefly described in this section.

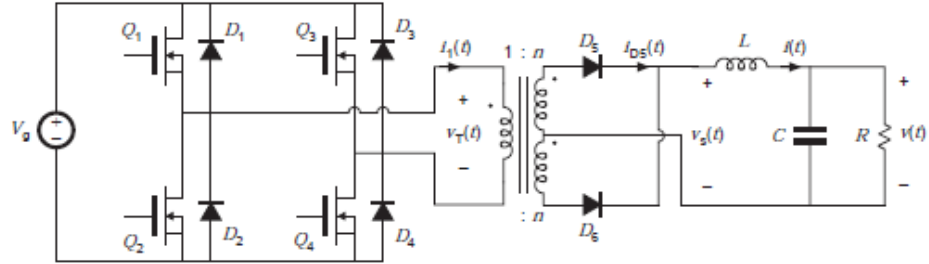


Figure 9. The full bridge transformer-isolated buck converter.

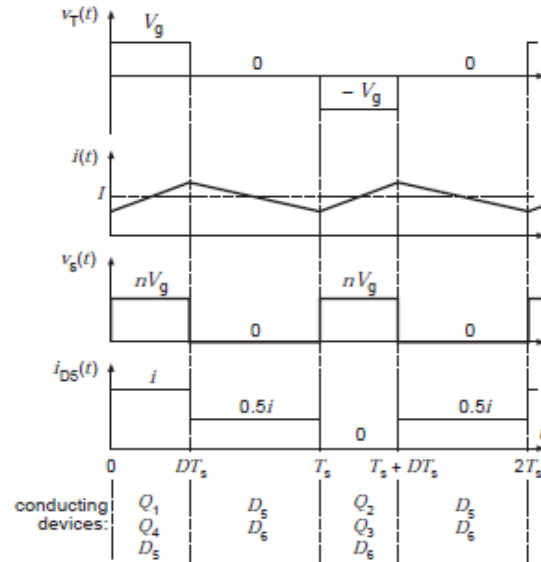


Figure 10. Waveforms of the full bridge circuit of Fig. 9.

Full-bridge buck-derived converter

The full-bridge transformer-isolated buck converter is sketched in Fig. 9. Typical waveforms are illustrated in Fig. 10.

The transformer primary winding is driven symmetrically, such that the net volt-seconds applied over two switching periods is equal to zero. During the first switching period, transistors Q_1 and Q_4 conduct

for time DT_s . The volt-seconds applied to the primary winding during this switching period are equal to $V_g DT_s$. During the following switching period, transistors Q_2 and Q_3 conduct for time DT_s , thereby applying $-V_g DT_s$ volt-seconds to the transformer primary winding. Over two switching periods, the net applied volt-seconds is equal to zero.

In practice, there exist small imbalances such as the small differences in the transistor forward voltage drops or transistor switching times, so that the average primary winding voltage is small but nonzero. This nonzero dc voltage can lead to transformer saturation and destruction of the converter. Transformer saturation under steady state conditions can be avoided by placing a capacitor in series with the transformer primary. Imbalances then induce a dc voltage component across the capacitor, rather than across the transformer primary. Another solution is the use of current programmed control; the series capacitor is then omitted.

By application of the principle of volt-second balance to the output filter inductor voltage, the dc load voltage can be shown to be

$$V = nDV_g \quad (11)$$

So, as in the buck converter, the output voltage can be controlled by adjustment of the transistor duty cycle D . An additional increase or decrease of the voltage V can be obtained via the physical transformer turns ratio n .

The full bridge configuration is typically used in switching power supplies at power levels of approximately 750 W or greater. At lower power levels, approaches such as the forward converter are preferred because of their lower parts count. Four transistors and their associated drive circuits are required. The utilization of the transformer is good, leading to small transformer size. The transformer operating frequency is one-half of the transistor switching frequency.

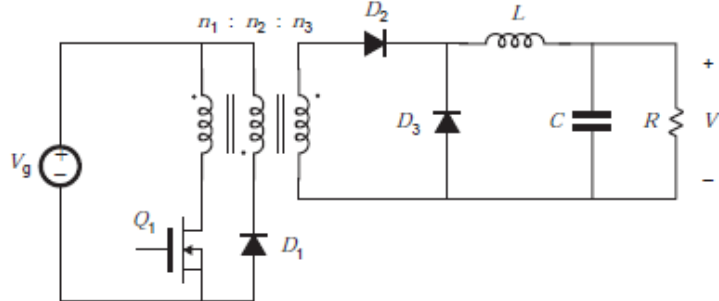


Figure 11. The forward converter, a single-transistor isolated buck converter.

Forward converter

The forward converter is illustrated in Fig. 11. This transformer-isolated converter is also based on the buck converter. It requires a single transistor, and therefore finds application at power levels lower than

those encountered in the full bridge circuit. The maximum transistor duty cycle is limited in value; for the common choice $n_1 = n_2$, the duty cycle is limited to the range $D < 0.5$.

The transformer is reset while transistor Q_1 is in the off state. While the transistor conducts, the input voltage V_g is applied across the transformer primary winding. This causes the transformer magnetizing current to increase. When transistor Q_1 turns off, the transformer magnetizing current forward-biases diode D_1 , and hence voltage $-V_g$ is applied to the second winding. This negative voltage causes the magnetizing current to decrease. When the magnetizing current reaches zero, diode D_1 turns off. Volt-second balance is maintained on the transformer windings provided that the magnetizing current reaches zero before the end of the switching period. It can be shown that this occurs when

$$D \leq \frac{1}{1 + \frac{n_2}{n_1}} \quad (12)$$

For the common choice $n_2 = n_1$, this expression reduces to

$$D \leq \frac{1}{2} \quad (13)$$

Hence, the maximum duty cycle is limited. If this limit is violated, then the transistor off time is insufficient to reset the transformer. There will then be a net increase in the transformer magnetizing current over each switching period, and the transformer will eventually saturate.

The converter output voltage can be found by application of the principle of inductor volt-second balance to the output filter inductor L . The result is

$$V = \frac{n_3}{n_1} D V_g \quad (14)$$

This expression is subject to the constraint given in Eq. (12).

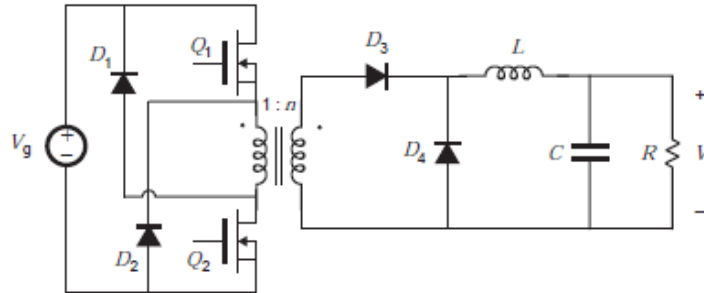


Figure 12. A two-transistor version of the forward converter.

A two-transistor version of the forward converter is illustrated in Fig. 12. Transistors Q_1 and Q_2 are controlled by the same gate drive signal, such that they conduct simultaneously. After the transistors turn off, the transformer magnetizing current forward-biases diodes D_1 and D_2 . This applies voltage $-V_g$ across the primary winding, thereby resetting the transformer. The duty cycle is again limited to $D < 0.5$. This converter has the advantage that the transistor peak blocking voltage is limited to V_g , and is clamped by diodes D_1 and D_2 . This circuit is quite popular in power supplies having 240 Vac inputs.

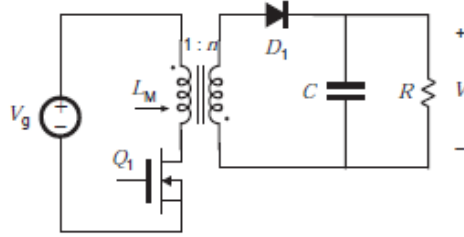


Figure 13. The flyback converter, a single-transistor isolated buck-boost converter.

Flyback converter

The flyback converter of Fig. 13 is based on the buck-boost converter. Although the two-winding magnetic device is represented using the same symbol as the transformer, a more descriptive name is “two-winding inductor.” This device is sometimes also called a “flyback transformer.” Unlike the ideal transformer, current does not flow simultaneously in both windings of the flyback transformer. Rather, the flyback transformer magnetizing inductance assumes the role of the inductor of the buck-boost converter. The magnetizing current is switched between the primary and secondary windings.

When transistor Q_1 conducts, diode D_1 is reverse-biased. The primary winding then functions as an inductor, connected to the input source V_g . Energy is stored in the magnetic field of the flyback transformer. When transistor Q_1 turns off, the current ceases to flow in the primary winding. The magnetizing current, referred to the secondary winding, now forward-biases diode D_1 . Energy stored in the magnetic field of the flyback transformer is then transferred to the dc load.

Application of the principle of inductor volt-second balance to the transformer primary winding leads to the following solution for the conversion ratio of the flyback converter:

$$M(D) = \frac{V}{V_g} = n \frac{D}{(1-D)} \quad (15)$$

Thus, the conversion ratio of the flyback converter is similar to that of the buck-boost converter, but with an added factor of n .

The flyback converter has traditionally been used in the high-voltage power supplies of televisions and computer monitors. It also finds widespread application in switching power supplies at the 50 W to 100 W power range. This converter has the advantage of very low parts count. Multiple outputs can be obtained using a minimum number of added elements: each auxiliary output requires only an additional winding, diode, and capacitor. However, in comparison with buck-derived transformer-isolated converters such as the full bridge and forward circuits, the flyback converter has the disadvantage of poor cross regulation.

Switch implementation

The switch network realization of Fig. 3 employs *single quadrant* switches. Each semiconductor element is able to conduct current of only one polarity in the on state, and block voltage of one polarity in the off state. This implies that, for proper functioning of the switch network, the source voltage, load

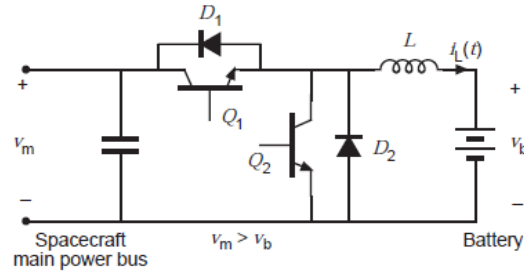


Figure 14. A buck converter with two-quadrant switches and bidirectional power flow. Spacecraft battery charger / discharger example.

voltage, and inductor current must all be positive. Consequently, the switch network allows the instantaneous power to flow in one direction only: from the source V_g towards to load.

Bidirectional (regenerative) power flow can be obtained with a current-bidirectional *two quadrant* realization of the switch network. An example is illustrated in Fig. 14, in which a dc-dc converter interfaces batteries to the main dc power bus of a spacecraft. The anti-parallel-connected transistors and diodes form current-bidirectional switches. Transistor Q_2 is driven with the complement of the Q_1 drive signal, such that Q_2 is off when Q_1 is on, and vice-versa. To charge the battery, the inductor current $i_L(t)$ is positive and flows through transistor Q_1 and diode D_2 . To discharge the battery, the current $i_L(t)$ reverses polarity, and flows through transistor Q_2 and diode D_1 . In both cases, the battery voltage is less than the main dc bus voltage. The magnitude and polarity of the battery current can be controlled via adjustment of the duty cycle D .

Switching loss imposes an upper limit on the switching frequencies of practical converters. During the switching transitions, the transistor voltage and current are simultaneously large. In consequence, the transistor experiences high instantaneous power loss. This can lead to significant average power loss, even though the switching transitions are short in duration. Switching loss causes the converter efficiency to decrease as the switching frequency is increased.

Several mechanisms lead to switching loss. Significant energy can be lost during the slow switching times of minority-carrier semiconductor devices such as BJTs, IGBTs, and thyristors. The diode reverse recovery process induces substantial additional energy loss in the transistor during the transistor turn-on transition. The energy stored in the semiconductor output capacitances is dissipated during the transistor turn-on transition. Energy stored in transformer leakage inductances and other stray inductances is usually dissipated by the transistor during the turn-off transition. The total switching loss is equal to the sum of the energy losses that arise via these mechanisms, multiplied by the switching frequency.

Small-signal ac modeling

To design the control system of a converter, it is necessary to model the converter dynamic behavior. In particular, it is of interest to determine how variations in the power input voltage $v_g(t)$, the load current, and the duty cycle $d(t)$ affect the output voltage. Unfortunately, understanding of converter dynamic behavior is hampered by the nonlinear time-varying nature of the switching and pulse-width modulation process. These difficulties can be overcome through the use of waveform averaging and small-signal modeling techniques [6-12]. A well-known converter modeling technique known as *state-space averaging* is briefly described here. Results for several basic converters are listed in tabular form.

State-space averaging

The state-space averaging technique generates the low-frequency small-signal ac equations of PWM dc-dc converters. Converter transfer functions and equivalent circuit models can be obtained.

The converter contains independent state variables such as inductor currents and capacitor voltages, that form the state vector $\mathbf{x}(t)$, and the converter is driven by independent sources that form the input vector $\mathbf{u}(t)$. The output vector $\mathbf{y}(t)$ contains dependent signals of interest. During the first subinterval, when the switches are in position 1 for time dT_s , the converter reduces to a linear circuit whose equations can be written in the following state-space form:

$$\begin{aligned}\frac{d\mathbf{x}(t)}{dt} &= \mathbf{A}_1\mathbf{x}(t) + \mathbf{B}_1\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C}_1\mathbf{x}(t) + \mathbf{E}_1\mathbf{u}(t)\end{aligned}\quad (16)$$

The matrices $\mathbf{A}_1$, $\mathbf{B}_1$, $\mathbf{C}_1$, and $\mathbf{E}_1$ describe the network connections during the first subinterval. The duty cycle $d(t)$ may now be a time-varying quantity. During the second subinterval, the converter reduces to another linear circuit, whose state space equations are

$$\begin{aligned}\frac{d\mathbf{x}(t)}{dt} &= \mathbf{A}_2\mathbf{x}(t) + \mathbf{B}_2\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C}_2\mathbf{x}(t) + \mathbf{E}_2\mathbf{u}(t)\end{aligned}\quad (17)$$

The matrices $\mathbf{A}_2$, $\mathbf{B}_2$, $\mathbf{C}_2$, and $\mathbf{E}_2$ describe the network connections during the second subinterval, of length $(1-d)T_s$.

It is assumed that the natural frequencies of the converter network are much smaller than the switching frequency. This assumption coincides with the small ripple approximation, and is usually satisfied in well-designed converters. It allows the high-frequency switching harmonics to be removed by an averaging process. In addition, the waveforms are linearized about a dc quiescent operating point. The converter waveforms are expressed as quiescent values plus small ac variations, as follows:

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{X} + \hat{\mathbf{x}}(t) \\ \mathbf{u}(t) &= \mathbf{U} + \hat{\mathbf{u}}(t) \\ \mathbf{y}(t) &= \mathbf{Y} + \hat{\mathbf{y}}(t) \\ d(t) &= D + \hat{d}(t)\end{aligned}\quad (18)$$

This small-signal linearization is justified provided that

$$\begin{aligned}
\|X\| &\gg \|x(t)\| \\
\|U\| &\gg \|u(t)\| \\
\|Y\| &\gg \|y(t)\| \\
D &\gg |\hat{d}(t)|
\end{aligned} \tag{19}$$

where $\|x\|$ represents the norm of vector x .

The state-space averaged model that describes the quiescent converter waveforms is

$$\begin{aligned}
0 &= AX + BU \\
Y &= CX + EU
\end{aligned} \tag{20}$$

where the averaged state matrices are

$$\begin{aligned}
A &= DA_1 + (1-D)A_2 \\
B &= DB_1 + (1-D)B_2 \\
C &= DC_1 + (1-D)C_2 \\
E &= DE_1 + (1-D)E_2
\end{aligned} \tag{21}$$

The steady-state solution of the converter is

$$\begin{aligned}
X &= -A^{-1}BU \\
Y &= (-CA^{-1}B + E)U
\end{aligned} \tag{22}$$

The state equations of the small-signal ac model are

$$\begin{aligned}
\frac{d\hat{x}(t)}{dt} &= A\hat{x}(t) + B\hat{u}(t) + \left[(A_1 - A_2)X + (B_1 - B_2)U \right] \hat{d}(t) \\
\hat{y}(t) &= C\hat{x}(t) + E\hat{u}(t) + \left[(C_1 - C_2)X + (E_1 - E_2)U \right] \hat{d}(t)
\end{aligned} \tag{23}$$

These equations describe how small ac variations in the input vector and duty cycle excite variations in the state and output vectors.

Canonical model

Equivalent circuit models of dc-dc converters can be constructed using the state-space averaged equations (20) and (23). Since all PWM dc-dc converters perform similar basic functions, one finds that the equivalent circuit models have the same form. Consequently, the *canonical circuit model* of Fig. 15 can represent the physical properties of PWM dc-dc converters.

The primary function of a dc-dc converter is the transformation of dc voltage and current levels, ideally with 100% efficiency. This function is represented in the model by an ideal *dc transformer*, denoted by a transformer symbol having a solid horizontal line. The dc transformer model has an effective turns ratio equal to the quiescent conversion ratio $M(D)$. It obeys all of the usual properties of transformers, except that it can pass dc voltages and currents. Although conventional magnetic-core transformers cannot pass dc voltages, we are nonetheless free to define an ideal dc transformer symbol; use of this symbol in modeling dc-dc converter properties is justified because its predictions are correct.

Small ac variations in the source voltage $v_g(t)$ are also transformed by the conversion ratio $M(D)$. Hence, a sinusoidal line is added to the dc transformer symbol, to denote that it also correctly represents how small-signal ac variations pass through the converter.

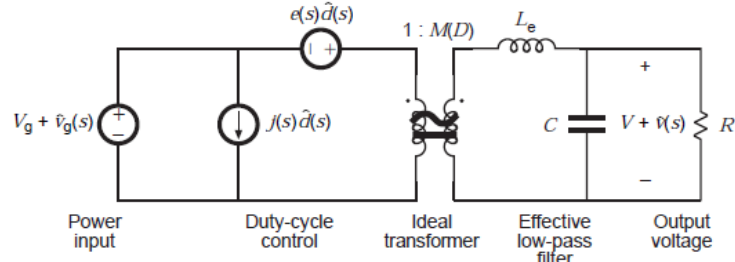


Figure 15. The canonical model: a small-signal equivalent circuit that models dc-dc converter dynamics and transfer functions.

Small ac variations in the duty cycle $d(t)$ excite ac variations in the converter voltages and currents. This is modeled by the $e(s)d$ and $j(s)d$ generators of Fig. 15. In general, both a current source and a voltage source are required.

The converter inductors and capacitors, necessary to low-pass filter the switching harmonics, also low-pass filter ac variations. The canonical model therefore contains an effective low-pass filter. Figure 15 illustrates the two-pole low-pass filter of the buck, boost, and buck-boost converters; complex converters having additional inductors and capacitors, such as the Cuk and SEPIC, contain correspondingly complex effective low-pass filters. The element values in the effective low-pass filter do not necessarily coincide with the physical element values in the converter. In general, the element values, transfer function, and terminal impedances of the effective low-pass filter can vary with quiescent operating point.

Canonical model parameters for the ideal buck, boost, and buck-boost converters are listed in Table 1. Transformer isolated versions of the buck, boost, and buck-boost converters, such as the full bridge, forward, and flyback converters, can also be modeled using the equivalent circuit of Fig. 15 and the parameters of Table 1; one must then correctly account for the transformer turns ratio by referring all quantities to the transformer secondary.

Table 1. Canonical model parameters for several basic converters

Converter	$M(D)$	L_e	$e(s)$	$j(s)$
Buck	D	L	$\frac{V}{D^2}$	$\frac{V}{R}$
Boost	$\frac{1}{(1-D)}$	$\frac{L}{(1-D)^2}$	$V \left(1 - \frac{sL}{(1-D)^2 R} \right)$	$\frac{V}{(1-D)^2 R}$
Buck-boost	$-\frac{D}{(1-D)}$	$\frac{L}{(1-D)^2}$	$-\frac{V}{D^2} \left(1 - \frac{sDL}{(1-D)^2 R} \right)$	$-\frac{V}{(1-D)^2 R}$

Small-signal transfer functions of the buck, boost, and buck-boost converters

The canonical circuit model of Fig. 15 can be solved using conventional linear circuit analysis techniques, to find quantities of interest such as the small-signal control-to-output and line-to-output transfer functions. The control-to-output transfer function $G_d(s)$ is the transfer function from $d(s)$ to $v(s)$, with $v_g(s)$ set to zero. The line-to-output transfer function $G_g(s)$ is the transfer function from $v_g(s)$ to $v(s)$, with $d(s)$ set to zero. For the buck, boost, and buck-boost converters, these transfer functions can be written in the following forms

$$G_d(s) = G_{d0} \frac{\left(1 - \frac{s}{\omega_z}\right)}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2} \quad (24)$$

$$G_g(s) = G_{g0} \frac{1}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2} \quad (25)$$

Expressions for the parameters of Eqs. (24) and (25) are listed in Table 2. The boost and buck-boost converters exhibit control-to-output transfer functions containing two poles and one right half-plane (nonminimum phase) zero. The buck converter $G_d(s)$ exhibits two poles but no zero. The line-to-output transfer functions of all three converters contain two poles and no zeroes.

The results of Table 2 can be applied to transformer-isolated versions of the buck, boost, and buck-boost converters, by referring all element values to the transformer secondary side. Equation (25) must also be multiplied by the transformer turns ratio.

The control systems of boost and buck-boost converters tend to be destabilized by the presence of the right-half plane (RHP) zero in the control-to-output transfer function. This occurs because, during a transient, the phase lag of the RHP zero causes the output to initially change in the wrong direction. When a RHP zero is present, it is difficult to obtain an adequate phase margin in conventional single-loop feedback systems having wide bandwidth. Prediction of the RHP zero, and the consequent explanation of why the feedback loops controlling continuous conduction mode boost and buck-boost converters tend to oscillate, was one of the early successes of averaged converter modeling and state-space averaging.

Table 2 Small-signal transfer function parameters for basic dc-dc converters

Converter	G_{g0}	G_{d0}	ω_0	Q	ω_z
Buck	D	$\frac{V}{D}$	$\frac{1}{\sqrt{LC}}$	$R \sqrt{\frac{C}{L}}$	∞
Boost	$\frac{1}{(1-D)}$	$\frac{V}{(1-D)}$	$\frac{(1-D)}{\sqrt{LC}}$	$(1-D)R \sqrt{\frac{C}{L}}$	$\frac{(1-D)^2 R}{L}$
Buck-Boost	$-\frac{D}{(1-D)}$	$\frac{V}{D(1-D)^2}$	$\frac{(1-D)}{\sqrt{LC}}$	$(1-D)R \sqrt{\frac{C}{L}}$	$\frac{(1-D)^2 R}{DL}$

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Conference on Electronics, Telecommunications and Computers – CETC 2013

DC-DC buck converter with reduced impact

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Abstract

One of the most constraints in DC-DC converter is the supply bouncing caused by the fast switching of power MOSFET. The purpose of this paper is to present a DC-DC buck converter using slew-rate modulator to increase the slew times of the PWM output instead the conventional output stage in order to minimize the bouncing. It does not constitutes a trivial task. This paper presents preliminary results of supply bouncing simulations made with a Slew-Rate Modulator. These simulation results show a very good agreement with the expected results.

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Peer-review under responsibility of ISEL – Instituto Superior de Engenharia de Lisboa, Lisbon, PORTUGAL.

Keywords: Supply bouncing; MOSFET power transistor; Buck converter; Slew-rate modulation; CMOS technology.

1. Introduction

Nowadays, the most important electronic devices that sharing the day-to-day of society, such as cellular phones, tablets and laptops are supplied by batteries and it require a very high autonomy. They have a lot of electronic circuits that should be supplied with different voltage levels, which is created the necessity to have circuitry capable of managing and distributing the voltages necessary for the proper operation of the electronic device [1].

Regarding the current scenario, with respect to the requirements of the new electronic devices, there is a growing need for improving the overall efficiency of a system through the development of these management blocks and power distribution that allow the minimization of losses. This is accomplished using topologies of DC-DC converters and their control schemes, so its integration in CMOS technology brings new challenges [1]. DC-DC

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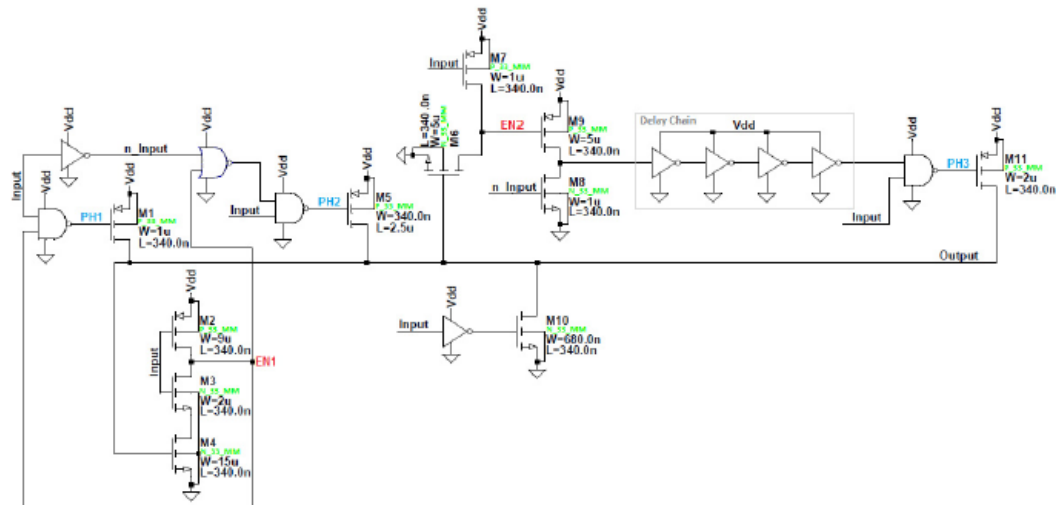


Figure 1. Schematic of the triple-phase slew rate modulator [4].

converters can be realized as linear, switched-capacitor, and inductive converters that temporarily store the energy transferred between the input source and the load.

DC-DC converters have been developed using increasingly improved methods makes them as a whole life support system, which must exhibit high power efficiency, high performance, together with a contribution significant reduction in area that can reduce overall system cost. In addition to the DC-DC converters allow generate multiple voltage levels from a single power source, they are used to attenuate ripples regardless the change in load current or input voltage [6]. Typically, they use pulse width modulation, PWM, to control one or more transistors (depending on the topology used). The switching of the power MOSFET controlled by the PWM usually causes supply bouncing. This kind of phenomenon will cause noise in remaining circuits. It can be reduced using a slew-rate modulation that allows increasing of slew times [7].

The main objective of this paper is to present a solution to reduce supply bouncing created by the transistor switching of power stage of CMOS inductor-based switch-mode DC-DC converters, based on the slew rate modulation of the signal which attacks the power gates. The slew rate modulator is presented in Section 2, including a timing analysis of the output signal provided by this modulator. The power loss model based in [1], [2], [3] is defined in Section 3. In Section 4 is showed an analysis of the efficiency of the converter versus the design parameters, and circuit simulation results for a 3.3–1.8V, 30mA buck converter working at a switching frequency of 100MHz, designed with the UMC 180nm CMOS technology.

2. Reducing supply bounce in DC-DC converters

The proposed work consists on a study of power electronics switched circuit topologies, followed for the development of a power circuit for the DC-DC buck converter with a technique to minimize the bouncing effect and based in the design methodology for DC-DC converters presented in [2], [3]. The main objective is the study and development of a circuit capable to minimize the supply bouncing caused by the switching of the power transistors. The idea presented by H. Jiao and V. Kursun in [4], [5] related activation and deactivation within blocks of microprocessors, as a way to reduce consumption, can be applied to switched converters. It consists in slew-rate modulation technique divide in three phases. The inclusion of a circuitry in addition to the non-overlap circuit which minimizes this effect, it has very high importance as way to reduce the impact of the introduction of this type of converters in a system on a chip, SoC.

Figure 1 shows the triple-phase slew rate modulator presented in [4], where it's possible identify the three transistors M_1 , M_5 and M_{11} , which are used for tuning the slew rate of input signal during reactivation events that occur in three distinct phases. These transistors are configured to the cut off in low level of the rising edge of the input signal. When there is a transition from low to V_{dd} , the transistor M_1 is turned on and the output signal starts to rise till when reach the threshold voltage defined by transistor M_4 . This corresponds to the phase 1, *PH1*, showed on the transient analysis illustrated in Figure 2. This simulation was performed for an input signal with period of 20ns.

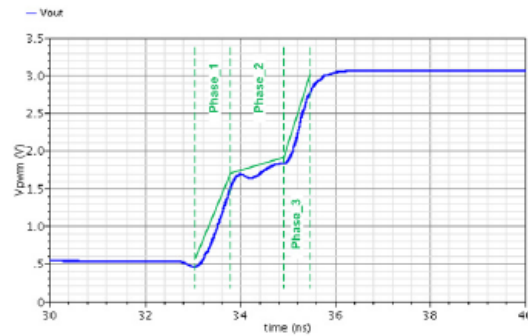


Figure 2. Transient analysis detailed waveform of output signal of the slew-rate control output stage.

After that, the M_1 is placed to the cut off and the phase 2, *PH2*, is started with M_5 turned on. In this phase, the slew rate of output signal is reduced as it is possible to observe in Figure 2. The delay chain presented in Figure 1 allows defining the time of *PH2*. The end of delay chain marks the third phase, *PH3*. In this last phase, the transistor M_{11} is turned on and the slew rate of output signal is raised faster toward V_{dd} .

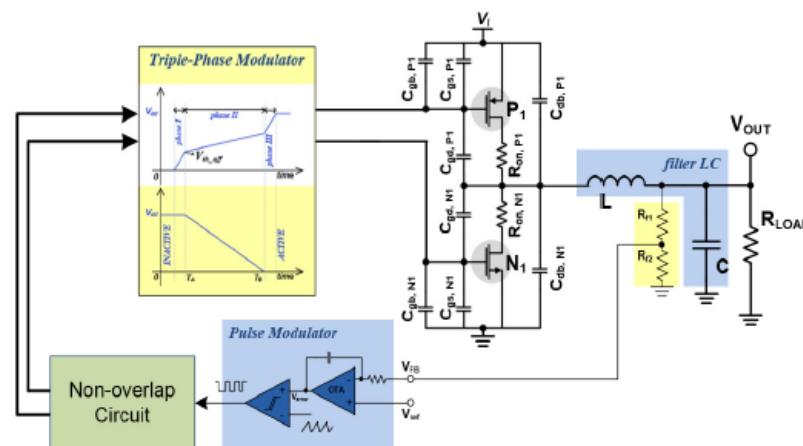


Figure 3. Circuit model of the power stage of a buck DC-DC converter, including parasitic impedances.

3. Power loss model

There are some approaches that show a circuit model developed to analyze the efficiency in function of the frequency [1], [2], [3]. The proposed circuit model for the parasitic impedances of the power stage of buck converter

is presented in Figure 3, and it is based on that works. However, it will be not considered the inclusion of gate drivers as mentioned, with the parasitic resistors and capacitors, because they cannot be used after the slew-rate modulator. This model will be centered just on the study of the dissipated energy due to conduction and switching losses in the power stage.

Assuming the conduction losses in the power stage are proportional to the conduction resistance of power MOSFET, $P_{cond} \propto R_{on} \times I_{rms}^2$, and switching losses are proportional to the intrinsic capacitances, $P_{sw} \propto C_{int} \times f_s$, it's possible to create a relationship between the cutback of both type of losses. The conduction losses for the two power MOSFETs (P_1 and N_1 in Figure 3) can be determined by the follow equations (1) and (2), where $i_{rmsPMOS}$ and $i_{rmsNMOS}$ are the effective current that flows through the PMOS and NMOS transistors, respectively, R_{0PMOS} and R_{0NMOS} are the PMOS ON-resistance per unit length, and W_{P1} and W_{N1} are the transistors width in μm . D is the duty-cycle, I the output current, and Δi_L is the current ripple in the filter inductor.

$$E_{PMOS_{conduction}} = \frac{R_{0PMOS}}{W_{P1}} i_{rmsPMOS}^2, \quad E_{NMOS_{conduction}} = \frac{R_{0NMOS}}{W_{N1}} i_{rmsNMOS}^2 \quad (1)$$

where:

$$i_{rmsPMOS} = \sqrt{D \cdot \left(I^2 + \frac{\Delta i_L^2}{3} \right)}, \quad i_{rmsNMOS} = \sqrt{(1-D) \cdot \left(I^2 + \frac{\Delta i_L^2}{3} \right)} \quad (2)$$

To get a fundamental understanding of the switching losses of the power stage, it is necessary to analyse the parasitic capacitance disposal resulting from the MOSFET physical process used on the design. The switching dissipated energy for the power transistors P_1 and N_1 , which compose the power stage are given by the equations (3) and (4), where C_{gb0} , C_{gd0} , C_{gs0} and C_{db0} are the gate-bulk, gate-drain, gate-source overlay and drain-bulk capacitances per unit length, respectively.

$$E_{P1TOTALswitching} = (C_{gb0PMOS} + C_{gs0PMOS}) (V_I - V_{gp})^2 + 2C_{gd0PMOS} \left(-V_I V_{gp} + V_I^2 + \frac{V_{gp}^2}{2} \right) + 2C_{db0PMOS} V_I^2 \quad (3)$$

$$E_{N1TOTALswitching} = (C_{gb0NMOS} + C_{gs0NMOS} + C_{gd0NMOS}) V_{gn}^2 + (C_{gd0NMOS} + C_{db0NMOS}) V_I^2 \quad (4)$$

The total power losses associated to P_1 and N_1 represented in Figure 3 are obtained as follows:

$$P_{P1TOTAL} = \frac{R_{0PMOS}}{W_{P1}} i_{rmsPMOS}^2 + W_{P1} E_{P1TOTALswitching} f_s \quad (5)$$

$$P_{N1TOTAL} = \frac{R_{0NMOS}}{W_{N1}} i_{rmsNMOS}^2 + W_{N1} E_{N1TOTALswitching} f_s \quad (6)$$

Choosing the transistors widths according to (7) will minimize the power losses (maximize efficiency) in the buck converter through an equilibrium between conduction and switching losses described on the Eqs. (5) and (6), where it's possible verify relationships between the losses, width for the power transistors and frequency.

$$W_{P1opt} = \sqrt{\frac{R_{0PMOS} i_{rmsPMOS}^2}{f_s E_{P1TOTALswitching}}}, \quad W_{N1opt} = \sqrt{\frac{R_{0NMOS} i_{rmsNMOS}^2}{f_s E_{N1TOTALswitching}}} \quad (7)$$

4. Simulated results

Figure 3 shows the circuit diagram of the DC-DC converter. The power circuit consists in a conventional synchronous buck configuration. Due to the focus on bouncing reduction, the prototype is implemented with a basic voltage mode PWM control and dominant pole compensation that generates corrective control signal to maintain the output voltage constant. These signals are modulated with triple-phase slew-rate modulator and it's applied to the power MOSFET. The output signal generated by slew-rate modulator is illustrated in Figure 2. The buck converter is being designed to be prototyped in a twin-well UMC 0.18um CMOS process. In this implementation, it is assumed that the buck converts 3.3V into 1.8V with 30mA of output current, output voltage ripple (ΔV_{out}) less than 1% of the output voltage, inductor current ripple (ΔI_L) below 50% of output current, and the switching frequency is 100MHz. The filter components are external.

$$L_{min} = \frac{V_{out}}{\Delta i_L \cdot f_s} (1-D) \quad (8)$$

$$\Delta V_{out} = \frac{V_{out} \cdot (1-D)}{8 \cdot L \cdot C \cdot f_s^2} \quad (9)$$

For the buck DC-DC converter, the required inductor and output capacitor are determined by the equations 8 and 9, respectively, where V_{out} is the output voltage, C is the capacitance value, and L_{min} is the inductor minimum value [1]. The values determined for the filtering capacitor is 1.67nF, and for the filtering inductor is 545nH. With the selected components, it's possible to calculate the system efficiency.

The dc-dc converter efficiency is in most cases a primary specification for any design target, because it is strongly dependent on the switching frequency, f_s . Regarding (3) and (4), it's verified that the values of L and C required to satisfy the target output voltage and current are reduced if f_s is increased. The converter efficiency is given by the (10), where the P_{out} is the output power and P_{buck} is the power losses.

$$\eta = 100 \times \frac{P_{out}}{P_{out} + P_{buck}} \quad (10)$$

Using the MATLAB tool and the equations on (7), it was possible generate 3-D graphs that correlate the power transistors width with the switching frequency and duty-cycle. The gate width of the switches P1 and N1 for target specifications are designed to be 660 um and 404 um, respectively, by equating the conduction loss and switching loss of the switches. However, regarding the following 3-D graphs (Figure 4 and Figure 5), it's observed that the transistors width decrease with the increasing of switching frequency.

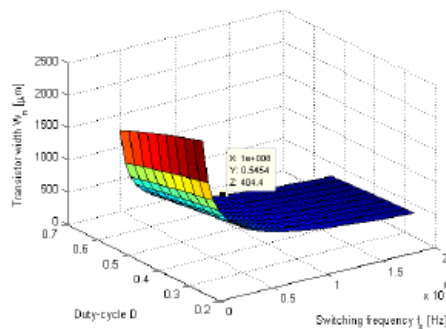


Figure 4. NMOS transistors width.

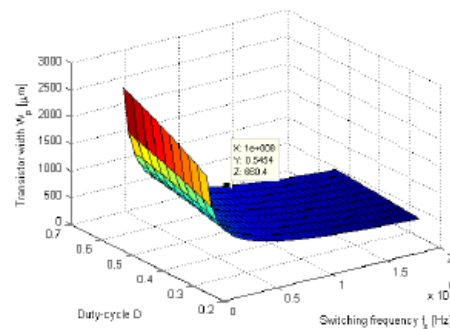


Figure 5. PMOS transistors width.

By the use of same analysis, in Figure 6 it's illustrated the buck converter efficiency according to the switching frequency and the duty-cycle. The calculated efficiency is 80.28% for a conversion from 3.3V to 1.8V with 30mA of output current, at a 100MHz switching frequency.

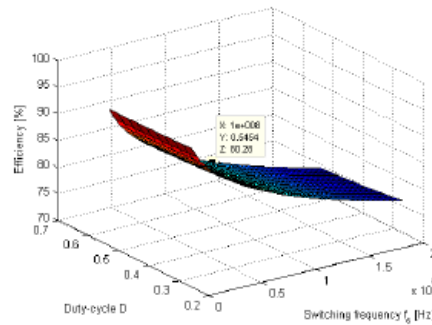


Figure 6. Buck converter efficiency.

Figure 6 shows the effect of different values of switching frequency on the converter's power efficiency where it's possible to see that the capacitive losses dominate at higher switching frequencies.

As can be seen the maximum efficiency is achieved when the frequency is about 5 MHz (more than 95% @ frequency 5 MHz) after this frequency the efficiency is decreased due to higher capacitive losses in the output power stage. From the same figure it's possible to conclude that the duty cycle variation on the buck converter have low influence on the buck DC-DC converter power efficiency.

An output voltage transient analysis of the buck converter is showed in Figure 7 belongs to the circuit presented in Figure 3, whereas can be seen the voltage drop across the load and the converter output stabilizes for 1.8V at 300ns. However, this value is not final, the work still under progress.

Figure 8 shows the output voltage ripple waveform of DC/DC buck converter circuit under following conditions (using type I compensation network); $V_{in} = 3.3V$, $V_{ref} = 1V$, $L = 545nH$, $C_L = 1.67nF$, $R_{load} = 60\Omega$ and switching frequency $f_s = 100MHz$. The simulated ripple of output voltage is 10mV. This value is less than the theoretical value of 18mV, which is 1% of output voltage. Figure 9 shows the transient response of inductor's peak-to-peak current ripple (Δi_L) and the load current which is equal to average of Δi_L , 30mA.

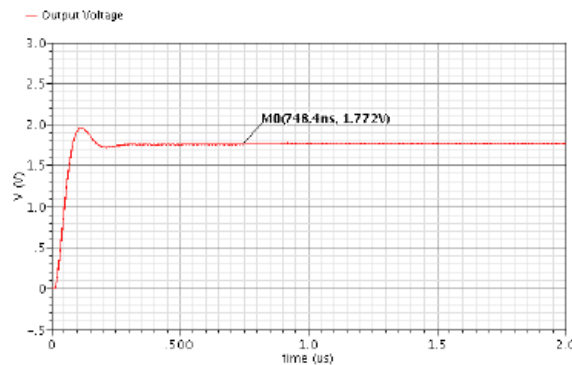


Figure 7. Output voltage transient analysis of the buck converter.

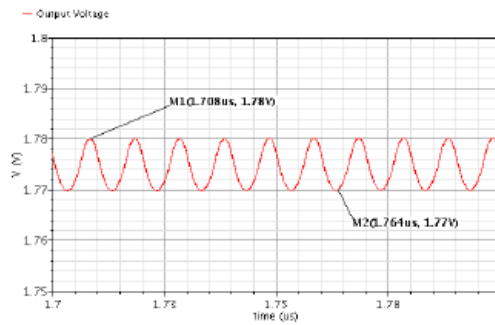


Figure 8. Output voltage ripple transient analysis of the buck converter.

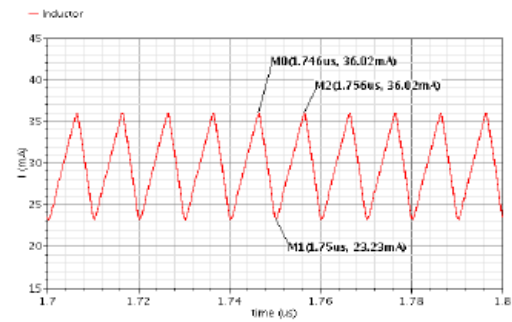


Figure 9. Inductor's peak-to-peak current ripple (Δi_L) transient analysis of the buck converter.

In Figure 10 it's represented the transient response result, validating the supply bouncing reduction from 800mV to 250mV by applying the slew rate modulation technique that corresponds to 65% of supply bouncing reduction, when compared with buck DC-DC converter with a conventional output stage. However, this value is a preliminary result because the work still under progress.

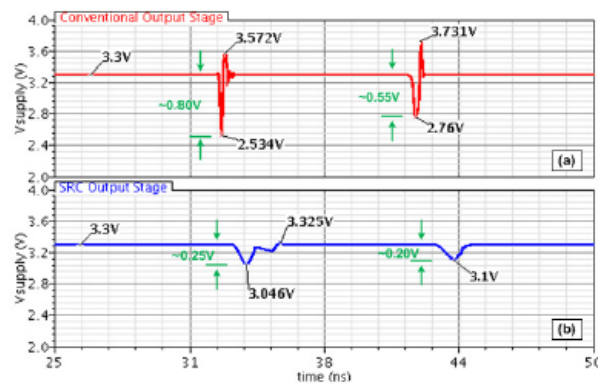


Figure 10. Transient analysis detailed waveforms of VDD with (a) the conventional output stage, and (b) the slew-rate control output stage, respectively.

Concerning to the total power delivered to the load is 53.2mW at maximum current, and the power supplied is 71mW, this means that the obtained efficiency is around the 75%.

5. Conclusions

In general, the supply bouncing caused by the fast switching of power MOSFET the converter constitutes a big problem during the design phase. This kind of issue will cause noise in all circuits that composes the DC-DC converter. To capitalise on the full advantage of the high frequency switching and low supply bouncing, a slew rate modulator integrated on the converter control has been proposed and simulation result shows that the proposed

topology has 65% of supply bouncing reduction when compared with the conventional topology composed by driver circuits. These preliminary results show a very good agreement with the supply bouncing reductions. The supply bouncing is significantly reduced by using the triple-phase slew-rate modulator stage instead of the conventional design. A prototype of the 100 MHz converter is being designed to be prototyped in a twin-well UMC 0.18µm CMOS process. Only the passive components like $L=545\text{nH}$ and $C=1.67\text{nF}$ will be off-chip. Simulation results show that, a maximum of 80% power efficiency is achieved, however this value is preliminary. By simulation, the converter has an output ripple is 10 mV with an output capacitor of 1.67nF.

Table 1 shows a comparison with a previously reported work [7]. This design [7] has similar input-output rating can be compared directly with this work. The proposed DC-DC buck converter shows a supply bouncing of 65% even at 100 MHz whereas [7] has only 40% at 500 kHz.

Table 1. Comparison between buck converters with supply bouncing solution.

Reference	[7] 2011	Simulations on this work
Switching frequency	500kHz	100MHz
Input range	3.3V	3.3V
Output voltage	1.8V	1.8V
Output current	700mA	30mA
Supply bouncing reduction	40%	65%
Efficiency	-	75%
Area	2.3mm ²	-
Process	0.35µm	0.18µm

The focus of this work is the supply bouncing reduction. All results presented on this paper are preliminary. This design work is ongoing.

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